

# PharmaChain — Single Source of Truth

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### Eliminating Fake Medicines from India Using Blockchain + AI/ML

**Document Purpose:** This is the **single source of truth** for the entire team. It covers every problem we are solving, every idea we have discussed, what is already solved, what government is working on, what is still open, and what we ship in V1. **Every team member must read this before writing a single line of code.**

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# §1 — Problem Landscape

**Legend for problem status tags:** -  **UNSOLVED** — No existing solution, we are building this -  **GOVT WORKING ON IT** — Government has started, not complete -  **PARTIALLY SOLVED** — Some solution exists but gaps remain -  **SOLVED** — Complete solution exists (by us or already in market) -  **IDEA** — Proposed, not yet committed to building

## 1.1 Core Fake Medicine Problems

| #    | Problem                                                          | Status                                                                                                        | Notes                                                                   |
|------|------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| P-01 | Fully counterfeit packs — manufactured with no legitimate origin |  <b>UNSOLVED</b>             | Our blockchain registry + manufacturer QR issuance closes this          |
| P-02 | Genuine QR code copied onto a fake/different box (relabeling)    |  <b>UNSOLVED</b>             | OCR physical-box verification in consumer app (§6.2)                    |
| P-03 | Genuine code resold after first sale (“already sold” code reuse) |  <b>UNSOLVED</b>             | State machine: one-time <b>Sold</b> flag with timestamp + location      |
| P-04 | Expired medicine sold as in-date                                 |  <b>PARTIALLY SOLVED</b>     | Expiry printed on pack; our system cross-checks against DB expiry       |
| P-05 | Recalled batch still on shelves — no real-time alert mechanism   |  <b>UNSOLVED</b>             | Push recall cascade to every shop holding the batch (§4.2.5)            |
| P-06 | Fake version of a legitimate drug (same name, wrong formulation) |  <b>UNSOLVED</b>           | OCR + batch-level formulation record verification                       |
| P-07 | Unregistered/unlicensed sellers distributing medicine            |  <b>GOVT WORKING ON IT</b> | KYC-gated onboarding; CDSCO licensing check                             |
| P-08 | Distributor/middleman as opaque black box in the supply chain    |  <b>UNSOLVED</b>           | Direct manufacturer-to-shop ordering (§3) eliminates need in many cases |
| P-09 | Shopkeepers have no incentive to scan every pack                 |  <b>UNSOLVED</b>           | Give them a free inventory system as a byproduct of scanning (§4.2)     |
| P-10 | Consumers don’t know they should verify medicine                 |  <b>UNSOLVED</b>           | Government awareness program + ASHA workers + POS signage (§9)          |
| P-11 | No real-time national counterfeit heatmap for regulators         |  <b>UNSOLVED</b>           | Regulator console with live district-level anomaly map (§4.4)           |

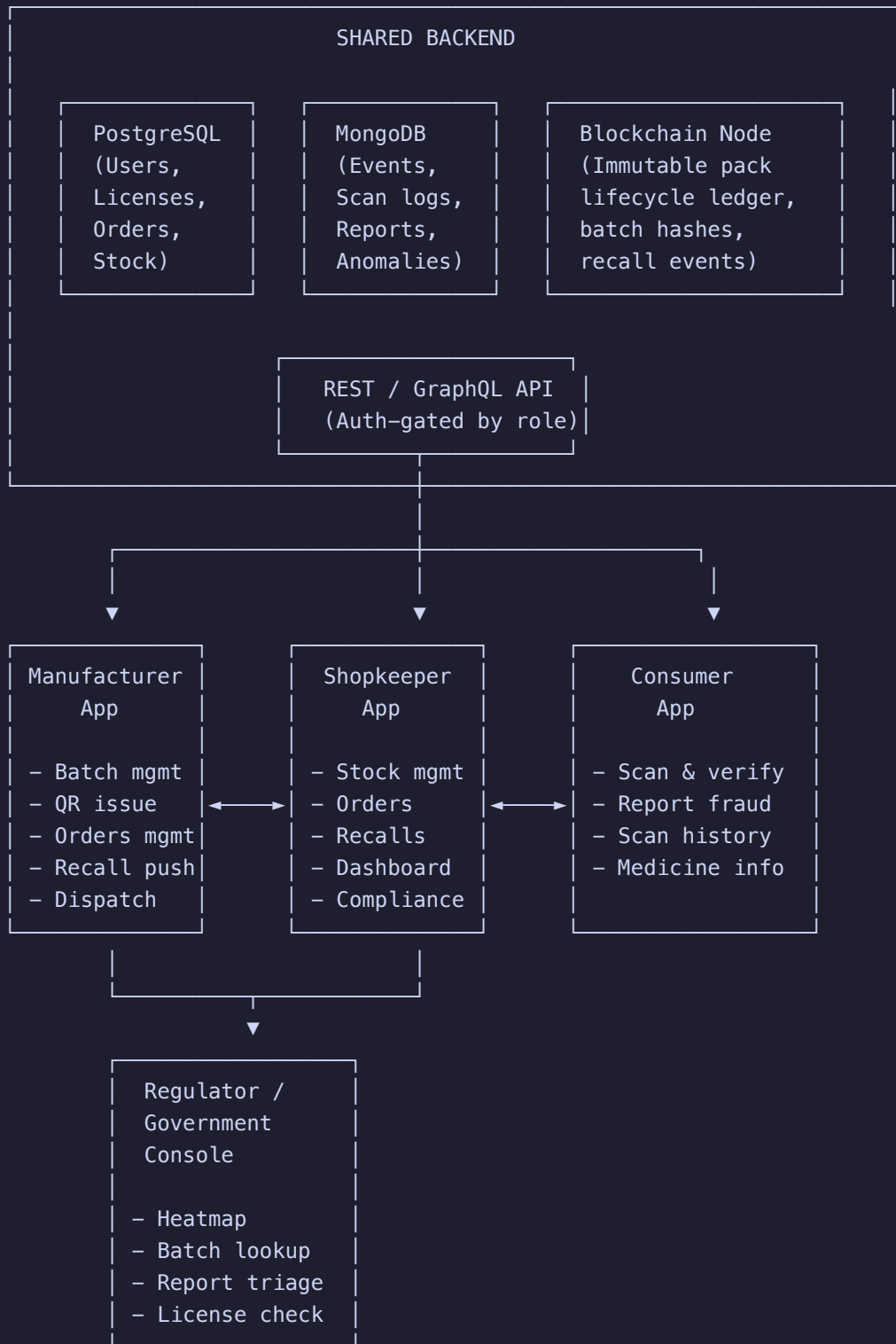
| #    | Problem                                                                | Status                                                                                                       | Notes                                                        |
|------|------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
| P-12 | Small pharmacies have no digital stock management<br>→ resist adoption |  <b>UNSOLVED</b>            | Shopkeeper app IS the inventory system (§4.2)                |
| P-13 | Fraud at distributor level<br>(diverting subsidized/govt medicines)    |  <b>GOVT WORKING ON IT</b>  | Our dispatch manifest + scan trail partially closes this     |
| P-14 | Medicines sold without prescriptions (scheduled drugs)                 |  <b>GOVT WORKING ON IT</b>  | Future integration with e-prescription                       |
| P-15 | No paperless purchase/sale register for pharmacies                     |  <b>UNSOLVED</b>            | Auto-generated register as byproduct of scan events (§4.2.4) |
| P-16 | Rural/low-literacy consumers cannot use smartphone-only app            |  <b>UNSOLVED</b>            | SMS/USSD/IVR fallback verification (§9)                      |
| P-17 | Jan Aushadhi medicines lack public-facing verification                 |  <b>GOVT WORKING ON IT</b> | Pilot our system there first — govt already owns this chain  |

## 1.2 Supply Chain Transparency Problems

| #    | Problem                                                               | Status                                                                                                      | Notes                                                             |
|------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|
| S-01 | No end-to-end traceability from manufacturer → consumer               |  <b>UNSOLVED</b>           | Core blockchain ledger solves this                                |
| S-02 | Batch recall takes days/weeks to reach all retailers                  |  <b>UNSOLVED</b>           | Real-time push recall cascade (§4.2.5, §4.3.2)                    |
| S-03 | Manufacturers have no visibility into where their products end up     |  <b>PARTIALLY SOLVED</b>   | Manufacturer anomaly dashboard scoped to their own batches (§4.3) |
| S-04 | No way for a shopkeeper to order directly from manufacturer           |  <b>UNSOLVED</b>           | <b>Direct Order Module</b> — our new feature (§3, §4.3.3)         |
| S-05 | Middlemen (distributors/C&F agents) add cost and opacity              |  <b>UNSOLVED</b>           | Direct ordering eliminates middleman for adopting shops (§3.2)    |
| S-06 | Shopkeeper cannot track which manufacturer/batch a delivery came from |  <b>PARTIALLY SOLVED</b> | Batch-linked AtShop scan + manufacturer dispatch manifest         |

## §2 — System Architecture — The Shared Backend Model

**Core design principle:** All applications (Manufacturer, Shopkeeper, Consumer, Regulator) are separate front-end apps but share a **common backend infrastructure**. They talk to each other via APIs over the shared DB — no app works in isolation.



## 2.1 Cross-App Communication Rules

- **Manufacturer writes, others read (mostly):** Manufacturer creates batches, issues QR tokens, manages recalls. All other apps consume this data.
- **Shopkeeper writes scan events:** **AtShop** and **Sold** events written by shopkeeper app are readable by manufacturer (anomaly view), consumer (verification), and regulator (audit).
- **Consumer writes reports:** Suspicious reports from consumer app feed directly into the regulator queue and anomaly pipeline.

- **Blockchain is append-only**: Every significant state change (batch created, pack sold, batch recalled) writes to the blockchain. Apps read from the DB for speed; blockchain is the audit anchor.
- **Role-based API access**: Same API, different endpoints unlocked based on `role` in JWT ( `MANUFACTURER` / `SHOPKEEPER` / `CONSUMER` / `REGULATOR` ).

## 2.2 Why Shared DB Matters

This is what makes the direct ordering feature possible (§3).

A manufacturer and shopkeeper exist in the **same user/entity registry**. When a shopkeeper places an order, the manufacturer sees it in their app. When the manufacturer ships, the shopkeeper's inventory is pre-populated. This works because they share the same backend — not because we built a separate B2B marketplace.

## §3 — The Supply Chain Flow — Eliminating the Middleman

### 3.1 Traditional Chain (What We Are Replacing / Complementing)

|                    |   |                  |   |                       |   |                   |   |          |
|--------------------|---|------------------|---|-----------------------|---|-------------------|---|----------|
| Manufacturer       | → | C&F Agent        | → | Stockist/Distributor  | → | Retailer/Pharmacy | → | Consumer |
|                    |   | ↑ (opacity here) |   | ↑ (opacity here)      |   |                   |   |          |
| No tracking        |   |                  |   | No real-time tracking |   |                   |   |          |
| No recall cascade  |   |                  |   | No recall cascade     |   |                   |   |          |
| Price markup ×2–3x |   |                  |   | Price markup ×1.5–2x  |   |                   |   |          |

**Problems with this chain:** - Every handoff is a potential point of diversion, duplication, or substitution - A recall at manufacturer level takes 1–3 weeks to reach all pharmacies - Shopkeeper does not know provenance of what they received - Consumer has zero visibility

### 3.2 New Chain — Direct Manufacturer-to-Shop (For Adopting Shops)

Manufacturer App —[Direct Order]→ Shopkeeper App

1. Shopkeeper places order (medicine, quantity, address)
2. Manufacturer reviews & confirms in their app
3. Manufacturer ships + logs dispatch on platform
4. Shopkeeper scans batch on arrival → AtShop event → stock auto-populated

**Benefits of direct ordering:** - Eliminates 1–2 levels of intermediaries → **cost saving for shopkeeper** - Every pack's origin is provably linked to the manufacturer who produced it - Recall cascade is instant and complete — manufacturer knows exactly which shops have which batches - Shopkeeper gets competitive pricing; manufacturer gets market intelligence on demand - Middle layer (C&F/distributor) is not eliminated by force — they become **optional** for shops that choose to go direct

### 3.3 Hybrid Chain — Traditional + Platform (Realistic V1)

For shops that still use distributors:

Manufacturer → Distributor → [Shopkeeper scans AtShop] → [Consumer scans Sold]

The platform does not require the distributor to be on-platform. The shopkeeper's **AtShop** scan is still the anchor. We miss the manufacturer→distributor leg traceability in this scenario — this is a known gap (see §10 open problems).

## §4 — Application Breakdown

### 4.1 Consumer App

**Goal:** Make it effortless for any consumer to verify any medicine in under 10 seconds.

#### 4.1.1 Scan & Verify — Full Status Table



| System finds...                                                           | Consumer sees                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Valid token, first Sold event, timestamp = today                          |  <b>Genuine — verified at time of purchase.</b><br>Manufacturer, batch, mfg/expiry, seller name & license shown.                                             |
| Valid token, Sold timestamp is weeks/months old or at a different shop    |  <b>This code was already marked sold on [date] at [shop/location].</b> May be a duplicate — use “Report suspicious.”                                        |
| Valid token, status = <b>Packaged</b> (never reached any registered shop) |  <b>This code has never checked into any registered shop.</b> Likely a cloned code — do not consume.                                                         |
| Token not found at all                                                    |  <b>This pack was never registered.</b> Strong signal of a fully counterfeit product.                                                                        |
| Token valid, physical box OCR mismatches the registered medicine/batch    |  <b>Box does not match what this code is registered for.</b><br>[Registered: medicine A / batch X] vs [Your box: medicine B / batch Y]. Relabeling red flag. |
| Status = <b>Recalled</b>                                                  |  <b>This batch has been recalled — do not consume.</b><br>[Reason if available.]                                                                             |
| Status = <b>ExpiredFlagged</b> or computed expiry has passed              |  <b>This pack is past its expiry date.</b>                                                                                                                 |
| Seller KYC not verified                                                   |  <b>Seller verification could not be confirmed.</b> Treat with caution.                                                                                    |

#### 4.1.2 OCR Physical-Box Verification (Catches Relabeling — P-02)

1. After QR scan, prompt: *“Take a photo of the medicine name and batch number on your box.”*
2. On-device OCR (Google ML Kit / Firebase ML) extracts text from the photo.
3. Compare extracted medicine name + batch number against the registered token record.
4. Mismatch → relabeling warning displayed. Match → normal Genuine display proceeds.

**Why this matters:** This is the only check that catches a genuine QR on a fake box. No amount of blockchain integrity catches physical-world fraud. This is our strongest technical differentiator.

#### 4.1.3 Other Consumer Features

- **My Medicines** — scan history; retroactive recall notifications if a batch they previously scanned gets recalled later
- **Report Suspicious** — structured tap-to-select categories: wrong medicine in box / seal tampered / price mismatch / adverse reaction / other
- **Seller legitimacy display** — shop name, license number, verification badge from KYC
- **Accessibility**: SMS/USSD fallback (text shortcode → get verdict by SMS), IVR toll-free call, regional language support

## 4.2 Shopkeeper App

**Goal:** Be the pharmacy's primary inventory system. Traceability is the free byproduct. *If the shopkeeper gets independent value from the app, they scan every pack.*

### 4.2.1 Live Stock Dashboard

Every **AtShop** and **Sold** scan auto-computes: - **On-shelf count** per medicine/batch = (AtShop scans) – (Sold + Recalled + ExpiredFlagged scans) - **Today / week / month** sold count by medicine and batch - **Value of current stock** — optional: enter cost price once per batch

**This replaces the paper register** most small Indian pharmacies still use. Zero manual counting required.

### 4.2.2 Expiry-Aging Report

Bucket current stock by days-to-expiry: -  < 30 days — one-tap “initiate return to distributor / manufacturer” -  30–90 days -  > 90 days

Helping the shopkeeper return near-expired stock before it becomes their loss is a **direct rupee saving** — the strongest adoption argument we have.

### 4.2.3 Reorder Signal

Simple velocity calculation: units sold per week per medicine → flag “reorder soon” when stock falls below ~2 weeks of run rate. Rolling average, no ML needed in V1.

### 4.2.4 Auto-Generated Compliance Register

Licensed pharmacies must maintain purchase/sale records for scheduled drugs. Since every pack's intake and sale is logged with timestamps, the app generates a downloadable PDF/Excel register for any date range at one tap.

**Pitch line:** “We did not build a compliance feature. We built an inventory tool that produces their compliance paperwork as a side effect.”

### 4.2.5 Recall Push Alerts

When a batch is flagged **Recalled**: 1. System queries all shops with on-shelf stock from that batch (visible from §4.2.1's running count) 2. Push notification to each shop: “Pull these exact packs. Batch [X], [N] packs on your shelf.” 3. Shopkeeper taps to confirm pull → **PullConfirmed** event logged → regulator sees action taken

### 4.2.6 Direct Ordering from Manufacturer

- Browse registered manufacturers and their available medicines/prices
- Place order: medicine, quantity, delivery address, requested date
- Track order status in-app (Confirmed / Shipped / Delivered)
- On delivery: scan packs → **AtShop** event auto-linked to the order → stock pre-populated

### 4.2.7 Recall Initiation by Shopkeeper

💡 *Feature idea:* If a shopkeeper notices a problem (wrong tablets in a bottle, seal damaged across multiple packs from the same batch), they can **flag a recall concern** to the manufacturer. The manufacturer reviews and either confirms (triggering full recall cascade) or dismisses with a reason. This creates a bottom-up recall signal that does not currently exist in the system.

#### 4.2.8 Multi-Outlet Roll-Up (*Stretch — Not V1*)

Manager view aggregating stock and sales across multiple pharmacy branches. Important for chains; not required for V1.

### 4.3 Manufacturer App

**Goal:** Full control of their product's lifecycle from production to shelf.

#### 4.3.1 Batch & QR Management

- Create batch: medicine name, formulation, batch number, mfg date, expiry date, quantity
- Generate cryptographically signed QR tokens for each pack (one token per pack, never reusable)
- Tokens are registered on blockchain at generation
- Print/export tokens for physical application to packs

#### 4.3.2 Recall Management

- One-tap recall trigger on any batch → cascades instantly to:
  - Every shopkeeper holding that batch (push notification, §4.2.5)
  - Every future consumer scan (shows Recalled status)
  - Regulator console (visible in triage queue)
- Track recall completion: see which shops have confirmed pulling the batch

#### 4.3.3 Direct Order Management (*New — Eliminates Middleman*)

- **Incoming Orders Dashboard:** All orders placed by shopkeepers across the platform
- Per order: shopkeeper details, license verification status, requested medicines, quantity, delivery address
- Confirm / modify / reject order with reason
- Mark order as shipped → auto-logs dispatch event linked to the batch on the platform
- View order history, fulfillment rate, and demand breakdown by shop / region / medicine

**What this creates:** A direct, traceable, auditable B2B channel between manufacturer and pharmacy — no distributor needed. For shops that adopt this, every pack is provably linked to its manufacturer from the moment it leaves the factory.

#### 4.3.4 Anomaly Dashboard (Scoped to Own Batches)

- Which batches are generating cloned QR reports, mismatch reports, or “not found” scans
- Geographic concentration — which region / distributor / shop is the locus of anomalies
- Trend over time per batch
- This gives manufacturers a **direct commercial incentive** to act: if their batch is being counterfeited in a specific region, they are losing sales there

### 4.3.5 Dispatch Manifest (for Traditional Distribution)

For orders going through distributors (not direct): “Batch X dispatched to Distributor Y on Date Z — Quantity N packs.” Single metadata field logged on the platform. Does not track custody through the distributor but creates an anchor point for the traceability chain.

## 4.4 Regulator / Government Console

**Goal:** Real-time intelligence dashboard for CDSCO, state drug authorities, enforcement teams.

### 4.4.1 National Counterfeit Heatmap

District-wise density of: - Cloned QR reports (valid token, sold multiple times) - Packaging mismatch reports (OCR vs. record) - “Not found” scans (no record → fully counterfeit) - Consumer “suspicious” structured reports

**Demo visual:** This map is the single strongest visual for the pitch deck and live demo.

### 4.4.2 Public Registry Lookup

Anyone can check: “Is manufacturer / shop license number X registered and verified on this platform?” — separate from scanning a specific pack.

### 4.4.3 Report Triage Queue

Consumer suspicious reports, sorted by: - Batch/shop concentration (many reports from same location = enforcement priority) - Report category (seal tampered > price mismatch in urgency) - Recency (recent clusters matter most)

### 4.4.4 Batch Audit Trail

Full chain-of-custody for any batch: Registered → Dispatched → AtShop (which shops, when) → Sold (how many, where) → Recalled (if applicable). Blockchain-anchored and tamper-proof.

### 4.4.5 License & KYC Management

- Onboard manufacturers and pharmacies (KYC gating)
- Verify / revoke licenses
- Flag entities for enforcement action

## §5 — Blockchain Architecture

### 5.1 Why Blockchain (and Why Not Just a Secured Database)

A secured database can be tampered with by the operator (us). Blockchain provides: - **Immutability:** A batch registration or recall event cannot be altered retroactively without detectable chain corruption - **Decentralization:** No single entity — including us — can unilaterally

alter history - **Auditability**: Any participant (manufacturer, regulator, researcher) can independently verify the chain - **Non-repudiation**: A manufacturer cannot deny they registered a batch; a shopkeeper cannot deny a sale event

## 5.2 What Goes on Chain vs. Database

| Event                                | On Chain?                                    | Reason                                  |
|--------------------------------------|----------------------------------------------|-----------------------------------------|
| Batch registered (manufacturer)      | ✓ Yes                                        | Immutable origin anchor                 |
| QR token issued per pack             | ✓ Yes (hash of token)                        | Proves token existed at issuance time   |
| Batch recalled                       | ✓ Yes                                        | Recall cannot be denied or backdated    |
| Pack: AtShop scan                    | ⚡ DB only (batch-anchored hash periodically) | High volume; full on-chain is expensive |
| Pack: Sold scan                      | ⚡ DB only (batch-anchored hash periodically) | Same reason                             |
| Suspicious report confirmed as fraud | ✓ Yes                                        | Evidence anchor                         |
| Order placed / fulfilled             | ⚡ DB only                                    | Business logic, not audit-critical      |

## 5.3 Chain Choice

- **Hyperledger Fabric** — permissioned, consortium model (CDSCO + large manufacturers as nodes)
- **Polygon (EVM)** — public, more decentralized, easier to demo
- For V1 / hackathon: local Hyperledger Fabric node or Polygon testnet
- For production: private/consortium chain shared with CDSCO, large manufacturers, state drug authorities

## 5.4 Token Structure

```
{
  "token_id": "UUID-v4",
  "batch_id": "BATCH-MFR-2025-X",
  "medicine_name": "Amoxicillin 500mg",
  "formulation": "Capsule",
  "manufacturer_id": "MFR-001",
  "mfg_date": "2025-01-15",
  "expiry_date": "2027-01-14",
  "status": "Packaged | AtShop | Sold | Recalled | ExpiredFlagged",
  "chain_hash": "0xABC...",
  "issued_at": "2025-01-15T10:00:00Z"
}
```

## §6 — AI/ML Layer

**Primary goal:** Use AI/ML to catch fraud that rule-based systems cannot — patterns invisible in any single transaction but visible across thousands of events.

### 6.1 Anomaly Detection — Statistical / ML Models

| Signal                         | What it catches                                           | Model type                |
|--------------------------------|-----------------------------------------------------------|---------------------------|
| Token scan velocity spike      | One QR code scanned from 10+ devices in 24h → cloned code | Threshold + Z-score       |
| Geographic impossibility       | Same pack scanned in Delhi and Mumbai within 1 hour       | Rule-based flag           |
| Shop AtShop/Sold ratio anomaly | Shop takes in 1000 packs, sells 50 → diversion signal     | Rolling average deviation |
| Batch complaint clustering     | Multiple “seal tampered” reports from same shop in 7 days | Clustering (DBSCAN)       |
| New shop, sudden high volume   | Account created last week, scanning 500 packs/day         | Isolation Forest          |
| Expiry mismatch                | Physical box expiry (OCR) differs from registered expiry  | String/date comparison    |
| Pricing anomaly                | Consumer reports price far above MRP for that batch       | Statistical outlier       |

### 6.2 OCR + Computer Vision — Physical Box Verification

- **On-device OCR:** Google ML Kit (free, no server call needed)
- Extracts: medicine name, batch number, expiry date, manufacturer name
- Cross-referenced against token record in DB
- Mismatch triggers relabeling fraud flag

**V1 scope:** Medicine name + batch number comparison. Expiry date and manufacturer name added in V1.1.

### 6.3 Demand Forecasting (V2 Idea)

💡 For shopkeeper: predict reorder needs based on historical scan velocity + seasonal patterns. 💡 For manufacturer: aggregate demand signal from all shops’ ordering patterns → production planning.

6.4 Adverse Event Pattern Detection (V2/V3 Idea)

💡 If consumers report adverse reactions for the same batch clustering in time → automatic signal to CDSCO / PvPI (Pharmacovigilance Programme of India).

6.5 Image-Based Counterfeit Detection (Aspirational)

💡 Train a vision model on genuine packaging images per medicine. Consumer photos → model flags visual anomalies vs. reference packaging. Requires a curated dataset from manufacturers — significant effort, not V1.

§7 — Government & Regulatory Landscape

7.1 ✅ What Government Has Already Done

| Initiative                          | What it does                                                | Our relationship                                                |
|-------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------------|
| CDSCO Drug Licensing                | Central licensing of manufacturers, importers, distributors | We integrate via KYC verification                               |
| Schedule H / H1 / H2 classification | Categorizes drugs requiring prescriptions                   | We enforce in our registry                                      |
| QR code mandate (Draft Rule, 2023)  | Mandates QR/barcode on Schedule H2 packs                    | <b>Our platform is the verification layer for this mandate</b>  |
| Jan Aushadhi Kendra network         | 18,600+ govt generic medicine stores                        | Our pilot deployment target (§9.4)                              |
| Mera Bill Mera Adhikaar             | Citizen incentive to upload GST invoices for cash prizes    | <b>Direct precedent for our consumer incentive model</b> (§9.3) |
| ASHA worker network                 | ~1M community health workers                                | Our behavioral rollout channel (§9.2)                           |
| Track and Trace System (draft)      | CDSCO's own serialization initiative (case/carton level)    | We complement — they track carton level, we track pack level    |
| State Drug Testing Labs             | Physical testing of samples                                 | We add the digital layer alongside                              |

## 7.2 🟡 What Government Is Working On (Incomplete)

| Initiative                                | Status                                               | Our gap-fill                                                     |
|-------------------------------------------|------------------------------------------------------|------------------------------------------------------------------|
| QR code mandate enforcement               | Notified; phased rollout from 2027; enforcement weak | We build the consumer verification habit before mandate kicks in |
| Digital pharmacy licensing centralization | In progress, fragmented across states                | We integrate available APIs; flag missing states in KYC flow     |
| National Drug Database                    | Being built by CDSCO                                 | We can feed anonymized scan data into this                       |
| e-Prescription system                     | State-by-state rollout, not national yet             | Future integration for scheduled drug sales                      |
| Pharmacovigilance reporting (PvPI)        | Exists but manual, low public awareness              | Our structured adverse-event reports can feed this pipeline      |

## 7.3 🔴 What Government Still Needs to Do (Gaps We Are Highlighting)

| Gap                                                    | Why it matters                                              | Our recommendation                                                                     |
|--------------------------------------------------------|-------------------------------------------------------------|----------------------------------------------------------------------------------------|
| No citizen-facing verification habit                   | QR mandate without consumer scanning = useless              | §9 awareness campaign is the fix                                                       |
| No real-time counterfeit intelligence for enforcers    | Raids happen on complaints, not data                        | Our regulator console (§4.4) provides live signal                                      |
| Distributor/C&F tracking is unregulated                | Black box in the supply chain                               | We partially bridge with dispatch manifests; advocate for GST-linked movement tracking |
| Rural accessibility of verification                    | Smartphone-only approach fails 50%+ of population           | SMS/USSD/IVR fallback (§4.1.3)                                                         |
| Manufacturer accountability after batch leaves factory | No mechanism to know which pharmacies have a recalled batch | Our shopkeeper inventory system + recall cascade solves this                           |

## §8 — Anomaly Detection Engine

### 8.1 Signal Sources

1. QR scan events (velocity, geography, timing)



2. Consumer structured reports (§4.1.3)
3. Shopkeeper inventory ratio anomalies (§4.2.1)
4. OCR mismatch reports (§4.1.2)
5. Order pattern anomalies (new account + unusually high order volume)

## 8.2 Fraud Patterns We Specifically Detect

| Pattern                                   | Detection method                                      | Severity                                                                                     |
|-------------------------------------------|-------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Cloned QR code                            | Same token Sold event > once; geo-impossible scans    |  Critical |
| Genuine QR on fake box                    | OCR mismatch between physical packaging and DB record |  Critical |
| Fully counterfeit pack                    | Token not found in registry at all                    |  Critical |
| Recalled batch still being sold           | <b>Sold</b> event on a <b>Recalled</b> batch          |  Critical |
| Diversion (stock taken in, never sold)    | High AtShop / low Sold ratio sustained over 30+ days  |  High     |
| Unauthorized seller                       | Shop not KYC-verified performing scan events          |  High     |
| Consumer adverse event reports clustering | Structured report + batch/location clustering         |  High   |
| Price above MRP                           | Consumer report                                       |  Medium |
| Batch complaint clustering at one shop    | DBSCAN on location + report type                      |  Medium |

## 8.3 Automated Response Actions

| Trigger                           | Automated action                                       | Human action required           |
|-----------------------------------|--------------------------------------------------------|---------------------------------|
| Cloned QR detected                | Flag token; alert manufacturer; log to regulator queue | Manufacturer + regulator review |
| OCR mismatch                      | Flag token; show consumer warning; log to anomaly DB   | Regulator review                |
| Recalled batch sold               | Block Sold status; alert shopkeeper + manufacturer     | Shopkeeper pull confirmation    |
| 5+ reports at same shop in 7 days | Escalate to regulator triage queue                     | Enforcement decision            |

## §9 — Government Awareness Program

### 9.1 Campaign Framing

One message, run like Swachh Bharat / Polio campaigns: **“Scan before you swallow.”** One QR scan, one clear verdict, every purchase. Make scanning as reflexive as checking an expiry date.

### 9.2 Channel Rollout

| Channel                                          | Who it reaches               | Cost                   |
|--------------------------------------------------|------------------------------|------------------------|
| POS signage at every licensed pharmacy           | All pharmacy customers       | Low (print)            |
| ASHA worker training module                      | Rural, low-literacy          | Low (existing network) |
| School health curriculum module                  | Next generation of consumers | Very low               |
| TV/Radio PSAs timed to QR mandate rollout (2027) | Mass market                  | Medium                 |
| Social media campaign                            | Urban, younger               | Low                    |

### 9.3 Incentive Mechanics — Mera Bill Mera Adhikaar Precedent

The government already proved this model works for GST compliance — citizens incentivized with cash prize draws to upload invoices built a verification habit at scale.

We propose: small cash / lottery rewards for consumer “report suspicious” submissions that are confirmed as fraud by the system.

**Judge answer:** “This is not a new idea — the government already proved it works for tax compliance. We are proposing the same mechanic for drug safety.”

### 9.4 Pilot Partner: Jan Aushadhi Kendras

Start where government **already owns the retail relationship**: - 18,600+ Jan Aushadhi Kendras across India (target: 25,000 by March 2027) - Government-run, government-trusted - Mandating scan flow + signage here = zero sales friction - This is our **credible, realistic V1 go-to-market** — not “convince 1M independent pharmacies on day one”

## §10 — Open Problems & Ideas Still to Solve

Flagged issues the team needs to think through. Nothing in this section is committed.

## 10.1 Unsolved — Distributor Black Box

**Problem:** Traditional supply chains route through distributors/C&F agents who are NOT on our platform. We have no visibility from manufacturer dispatch → shopkeeper receipt.

**Current mitigations:** Dispatch manifest at manufacturer; AtShop scan at shopkeeper.

**Gap:** If a distributor adulterates or substitutes medicines before delivery, we cannot detect it from scan events alone.

**Ideas to explore:** - Partner with large distributors to add an **AtDistributor** scan event - Use GST e-way bill data as a proxy for movement tracking - Incentivize distributors to onboard by offering them a logistics dashboard

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## 10.2 Unsolved — Manufacturer Itself as Bad Actor

**Problem:** What if a licensed manufacturer intentionally produces substandard/fake medicine?

**Current mitigations:** CDSCO licensing; state drug testing labs do periodic sampling.

**Gap:** Our system trusts that a registered manufacturer's registered batch is genuine. No in-platform mechanism to verify batch quality.

**Ideas:** - Integrate with CDSCO's drug testing results — if a batch fails testing, auto-flag all tokens from that batch - Anomaly signal: if a specific manufacturer's batches generate disproportionate adverse event reports, escalate automatically to regulator

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## 10.3 Unsolved — Physical QR Code Tampering

**Problem:** What if the physical QR sticker is removed from a genuine pack and placed on a fake pack?

**Current mitigation:** Tamper-evident packaging mandate exists for some drug categories.

**Gap:** We depend on the physical QR being unremovable / un-reproducible.

**Ideas:** - Push manufacturers to use QR codes printed directly on packaging (not sticker-applied) - Multi-factor check: QR code + batch number + expiry from OCR must all match simultaneously

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## 10.4 Partially Solved — Offline Areas

**Problem:** Rural pharmacies may lack reliable internet for real-time scans.

**Current mitigations:** SMS/USSD for consumers.

**Gap:** Shopkeeper offline scan queue that syncs when online?

**Ideas:** - Local event queue in app with batch sync when connectivity is available - Flag: scan events more than 24h delayed from their timestamp get an audit note

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### 10.5 💡 Idea — e-Prescription Integration

When e-prescriptions become standard, link sold packs to the prescription ID. Catches dispensing of controlled substances without prescriptions. Long-term integration goal.

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### 10.6 💡 Idea — Cold Chain Monitoring

Some medicines (vaccines, biologics) require cold chain. Temperature logs from IoT sensors at storage/transport points could be linked to batch records. A broken cold chain does not make a medicine fake but makes it ineffective — same harm to patient. V3 feature.

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### 10.7 💡 Idea — Consumer Medicine Interaction Checker

After verifying a medicine as genuine, offer: “Are you taking other medicines? Check for interactions.” Uses public pharmacopoeia data. High consumer value, zero security scope change.

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### 10.8 💡 Idea — Manufacturer Competitive Intelligence (Opt-In)

Aggregate anonymized sales velocity data from shopkeeper scans → show manufacturers which of their medicines are selling in which regions. Creates a monetization path and additional platform stickiness for manufacturers.

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## §11 — V1: What We Actually Ship

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**Rule:** Build = working demo. Describe = slide/mockup. These two categories must never blur.

11.1  Build for the Demo

| Feature                                                    | Why in V1                                       | Effort |
|------------------------------------------------------------|-------------------------------------------------|--------|
| Manufacturer: Batch creation + QR generation               | Core of the whole system                        | Medium |
| Manufacturer: Direct order management                      | New differentiator; middleman elimination story | Medium |
| Manufacturer: One-tap recall + cascade                     | High visual impact in live demo                 | Low    |
| Shopkeeper: AtShop + Sold scan events                      | Foundation of everything                        | Low    |
| Shopkeeper: Live stock dashboard (§4.2.1)                  | Primary adoption driver                         | Medium |
| Shopkeeper: Expiry-aging report (§4.2.2)                   | Direct rupee value to shopkeeper                | Low    |
| Shopkeeper: Direct order placement (§4.2.6)                | Ties to manufacturer's order management         | Medium |
| Shopkeeper: Recall push alert + confirm pull (§4.2.5)      | Strong demo moment                              | Low    |
| Consumer: Full scan + status display (§4.1.1)              | Core user-facing feature                        | Medium |
| Consumer: OCR physical-box mismatch check (§4.1.2)         | Biggest technical differentiator                | Medium |
| Consumer: Structured "report suspicious" (§4.1.3)          | Feeds anomaly pipeline                          | Low    |
| Regulator: Counterfeit heatmap (§4.4.1)                    | Best demo visual                                | Medium |
| Blockchain: Batch registration + recall events on-chain    | Anchors the "tamper-proof" claim                | Medium |
| AI/ML: Basic anomaly detection (cloned QR, velocity spike) | Makes "AI-powered" claim real                   | Medium |

## 11.2 Describe in Pitch — Do Not Build in V1

| Feature                                     | Where to mention                                                    |
|---------------------------------------------|---------------------------------------------------------------------|
| SMS/USSD/IVR consumer fallback              | Pitch deck — accessibility slide                                    |
| Multi-outlet shopkeeper roll-up             | Pitch deck — scale slide                                            |
| Reorder velocity analytics (ML)             | Pitch deck — future AI slide                                        |
| Demand forecasting for manufacturers        | Pitch deck — future AI slide                                        |
| e-Prescription integration                  | Pitch deck — government integration slide                           |
| Cold chain IoT integration                  | Pitch deck — future hardware slide                                  |
| ASHA worker training rollout                | Pitch deck — go-to-market slide                                     |
| School curriculum module                    | Pitch deck — long-term impact slide                                 |
| Consumer lottery/incentive mechanics        | Pitch deck — adoption strategy slide (cite Mera Bill Mera Adhikaar) |
| Distributor onboarding + AtDistributor scan | Pitch deck — known gap + roadmap                                    |
| Image-based packaging counterfeit detection | Pitch deck — V3 AI roadmap                                          |

## 11.3 V1 Non-Negotiables (Cannot Skip)

1. **Blockchain must be real** — at minimum a local Hyperledger Fabric node or Polygon testnet; not simulated in a regular DB
2. **OCR mismatch check must work live** — on a real phone camera; this is our key demo moment
3. **Shared backend** — manufacturer and shopkeeper apps must share the same DB so direct ordering works end-to-end
4. **Recall cascade must be live-demo-able** — recall a batch in the manufacturer app, instantly see the push notification on the shopkeeper device

## §12 — Judge Q&A Pre-Answers

| Judge asks...                                                            | Our answer                                                                                                                                                                                                   |
|--------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| “Why would a shopkeeper actually scan every pack?”                       | Because the same scan gives them a live stock count, expiry alerts, and their compliance register for free (§4.2). Traceability is the free byproduct.                                                       |
| “How do you catch a real QR on a fake box?”                              | OCR the physical packaging; compare medicine name + batch number against what the QR token actually points to (§4.1.2, §6.2). No DB integrity alone can catch physical-world fraud.                          |
| “Why would a manufacturer use this instead of their own system?”         | We give them a direct sales channel to retailers eliminating C&F margin, plus real-time intelligence on which of their batches are being counterfeited and where (§4.3.3, §4.3.4).                           |
| “Is this just an app nobody will actually use?”                          | We pilot at 18,600 Jan Aushadhi Kendras first — where adoption is a policy decision, not a sales problem (§9.4).                                                                                             |
| “Has anyone ever gotten citizens to adopt a verification app like this?” | Yes — Mera Bill Mera Adhikaar did exactly this for GST invoices using cash prize incentives. We are proposing the same mechanic for drug safety (§9.3).                                                      |
| “Is blockchain overkill? Why not just a database?”                       | A database we operate can be altered by us. Blockchain ensures that even we — and the government — cannot retroactively modify a batch registration or recall event. That is the point (§5.1).               |
| “What about distributors? You are missing a layer.”                      | Acknowledged gap (§10.1). We cover it with a manufacturer dispatch manifest. Direct ordering (§4.2.6) eliminates the distributor entirely for shops that adopt it. Distributor onboarding is on the roadmap. |
| “What if the manufacturer itself is corrupt?”                            | CDSCO testing labs handle quality certification. We integrate their flagged-batch data. Our anomaly engine also flags manufacturers whose batches generate disproportionate adverse reports (§10.2).         |
| “What about rural areas with no internet?”                               | SMS/USSD/IVR fallback for consumers. Offline scan queue for shopkeepers that syncs when connectivity returns (§10.4).                                                                                        |

| Judge asks...                                                | Our answer                                                                                                                                                                                              |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| “How is this different from CDSCO’s Track and Trace system?” | CDSCO tracks at the case/carton level for regulatory reporting. We track at the individual pack level with consumer-facing verification and a real-time fraud signal layer. We complement, not replace. |

 **The Main Goal — One Sentence**

**Use blockchain to make every medicine pack verifiably genuine at the point of manufacture, AI/ML to detect fraud patterns no single scan can reveal, and direct manufacturer-to-shop ordering to eliminate the middlemen who create opacity — so that by the time a consumer scans a pack, the answer “genuine” is one they can fully trust.**

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